Keywords: positional encoding • 2D-RoPE • in-context retrieval • transformer architecture

Frontier Language Models Struggle to Copy: Text Can Be Better Viewed in 2D

A Structural Failure in 1D Positional Encoding Prevents Exact String Copying—2D-RoPE Solves It at Training Scales up to 1.4B Parameters

By Haodong Wen, Yiran Zhang, Yingfa Chen, Kaifeng Lyu (Tsinghua University)

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Frontier language models solve complex multi-step reasoning yet consistently fail at a task any human child can perform: copying a string exactly. A new paper from Tsinghua University traces this failure to a structural flaw in standard 1D rotary positional encodings and proposes **2D-RoPE**, which organizes tokens into a two-dimensional grid so that copying reduces to zero-offset column retrieval. Shallow Transformers with 2D-RoPE achieve perfect copying on sequences hundreds of times longer than their training range.

AT A GLANCE

Problem: Standard 1D positional encodings force copying to rely on local-context shortcuts whose relative offsets change with sequence length, causing systematic failure on long strings.

Why it matters: Verbatim extraction from context—for quotes, code formatting, structured outputs—is foundational; a structural fix is more reliable than prompt engineering.

Method: 2D-RoPE assigns each token a row ID and column ID; copying at fixed column offset becomes a zero-relative-offset attention task.

Result: 6-layer Transformers with 2D-RoPE reach 99.7% exact copy accuracy at lengths 8× beyond training range; standard RoPE achieves 12%.

Evidence: Advantage holds in 1.4B pretraining experiments and on downstream verbatim extraction benchmarks.

The Big Picture

Large language models have surpassed humans on olympiad mathematics, bar exams, and competitive programming—but they cannot reliably copy a 1024-token string they just read. This asymmetry is not a quirk of training data; it reflects a structural mismatch between how attention mechanisms encode position and what the copy task requires.

The paper demonstrates the failure empirically on GPT-4o, Claude 3.7, and Gemini 2.0: exact-copy accuracy falls below 50% on 512-token strings and degrades further on repetitive inputs. The authors then derive a root cause from the mathematics of RoPE, and propose a fix that achieves perfect copying at lengths orders of magnitude beyond training.

Why Existing Approaches Fall Short

In Transformer attention with RoPE, the attention score between query position p and key position q depends on the relative offset $p - q$. Copying token x_k to output position y_k requires attending to exactly input position k. But as the output grows, the relative offset $k - y_k$ changes—so the model must infer k dynamically rather than attend to a fixed offset.

In practice, models learn to copy by matching local context: if a substring appears only once, the model retrieves it via pattern matching rather than positional address. This works for unique strings but breaks when strings repeat (the context match is ambiguous) or are very long (the attention score for the correct position degrades relative to spurious matches). Prompt-engineering workarounds ("copy exactly") improve but do not solve the problem because the architectural inductive bias remains.

Core Mechanism

2D-RoPE organizes a sequence of N tokens into a grid with R rows and $C = N/R$ columns. Token i receives:

- Row ID: $r_i = \lfloor i/C \rfloor$
- Column ID: $c_i = i \bmod C$

Rotary encoding is applied along each dimension independently. Under this view, copying is: for output position (r, c), retrieve the input token at the same column c in row r. The column offset between any output token and its corresponding input token is always zero—a fixed, length-independent offset that the model can reliably learn.

The key invariant: *the relative column offset for copying is 0, regardless of sequence length.* Standard 1D RoPE has no equivalent fixed offset.

Technical Formulation

Standard RoPE applies rotation $e^{ip\theta}$ at position p with frequency θ, giving attention score:

$$A_{pq} = \mathrm{Re}\left[q_p^* k_q \cdot e^{i(p-q)\theta}\right]$$

2D-RoPE splits the embedding dimension: half the heads attend to row position, half to column position:

$$A_{ij} = \mathrm{Re}\Big[q_i^* k_j \cdot e^{i[(r_i - r_j)\theta_R + (c_i - c_j)\theta_C]}\Big]$$

During a copy task, $r_i - r_j = 0$ (same row, input vs. output) and $c_i - c_j = 0$ (same column), so:

$$A_{ij}^{\text{copy}} = \mathrm{Re}[q_i^* k_j]$$

The attention score reduces to the inner product of query and key without any positional modulation—a fixed, position-independent signal that is identical at any sequence length.

Learning or Inference Procedure

Training proceeds identically to standard causal language modeling, replacing the 1D position index with the 2D (r, c) pair. No changes to the loss function, optimizer, or data pipeline are required. The grid dimensions R and C are treated as hyperparameters; the paper uses $C = 64$ for all experiments (so rows contain 64 tokens).

At inference, sequences longer than training length are handled by extending the row dimension: a 4096-token sequence with $C = 64$ requires $R = 64$ rows, while a 512-token training sequence requires only $R = 8$ rows. Column IDs remain bounded by C, so the column encoding generalizes trivially to longer sequences.

What the Guarantee Says

For a sequence of length N with column width C, 2D-RoPE guarantees that the copy attention signal is *position-independent* at offset $(0, 0)$ in row-column space. This is a structural, not statistical, guarantee: no amount of sequence length increase can alter the positional signal the model must attend to in order to copy correctly.

The guarantee does not cover sequences longer than $R_{\max} \cdot C$ (where $R_{\max}$ is the training row count), but row generalization is analogous to standard length generalization and can be addressed with existing techniques (ALiBi, YaRN).

Experimental Findings

Synthetic copy task (exact accuracy at length L, trained on $L = 512$):

Model	$L = 2048$	$L = 4096$
Standard RoPE (6L)	34.1	12.3
2D-RoPE (6L)	98.4	99.7
Standard RoPE (12L)	41.8	18.7
2D-RoPE (12L)	99.1	99.8

Repetitive strings (period 4, length 512, in-distribution): Standard RoPE: 7.3%. 2D-RoPE: 98.1%.

Pretraining at 1.4B parameters (100B tokens, same perplexity as baseline): 2D-RoPE models improve on verbatim extraction benchmarks by 14.2 points while matching or exceeding baseline on MMLU, HellaSwag, and ARC-Challenge.

Ablations and Interpretation

Grid aspect ratio: Varying $C \in \{16, 32, 64, 128\}$ shows $C = 64$ is optimal; very narrow grids ($C = 16$) reduce copy benefit by requiring more rows and increasing within-row attention competition.

Attention visualization: On copy tasks, 2D-RoPE heads in the column-position half show clean offset-0 attention spikes at the source token; standard RoPE models show diffuse attention over a ± 5 token window, confirming the local-context shortcut.

Frontier model analysis: Attention rollout on GPT-4o and Claude 3.7 during copy failures reveals the same diffuse-attention signature, confirming the failure mode is not specific to small models.

Column vs. row only: Using only column encoding (dropping row) degrades general language modeling perplexity by 0.4 bits while maintaining near-perfect copying—confirming that row and column encodings serve distinct roles.

Reference

Haodong Wen, Yiran Zhang, Yingfa Chen, Kaifeng Lyu. **Frontier Language Models Struggle to Copy: Text Can Be Better Viewed in 2D**. arXiv:2607.16072, July 15, 2026. https://arxiv.org/abs/2607.16072

Keywords: multimodal reasoning • reinforcement learning • vision-language models • geometry

MIRROR: Learning from the Other View for Multi-Modal Reasoning

Modality-Informed Reciprocal Reasoning Optimization Trains Vision-Language Models to Transfer Strengths Across Text, Diagram, and Combined Views of the Same Problem

By Wen Ye, Yuxiao Qu, Aviral Kumar, Xuezhe Ma (University of Southern California; Carnegie Mellon University)

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Vision-language models fail at visual reasoning even on geometry problems that have mathematically equivalent text, diagram, and combined views. MIRROR, from researchers at USC and CMU, exploits this inconsistency: rather than treating it as noise, the method identifies which view a model currently solves best and uses that signal to lift performance on the harder view through reciprocal cross-modal reward shaping during RL training. On ODA-Data, a new paired multimodal geometry benchmark, MIRROR improves image-dominant reasoning by 17 points over a strong baseline.

AT A GLANCE

Problem: VLMs solve the same geometry problem in text but fail on the diagram, revealing complementary reasoning gaps across modalities that standard post-training ignores.

Why it matters: Closing modality gaps enables VLMs to reason reliably regardless of how a problem is presented—critical for real-world deployment where input format is unpredictable.

Method: MIRROR evaluates each view per-problem, identifies the best-performing view, and shapes rewards on weaker views toward the best-view distribution via RL.

Result: +17pp on image-dominant and +9pp on combined view vs. GRPO baseline on ODA-Data geometry; consistent gains on three external geometry benchmarks.

Evidence: Ablating cross-modal shaping loses 9.4pp image accuracy; scrambled skills confirm content drives improvement.

The Big Picture

Vision-language models are post-trained on massive image-text pairs and achieve strong results on captioning, VQA, and chart understanding. Yet on mathematical visual reasoning—where the model must interpret a geometric diagram and apply deductive inference—performance lags substantially behind text-only reasoning.

The standard explanation is that vision encoding is weaker than language encoding. MIRROR's key insight is that this framing misses a more specific phenomenon: the *same problem* in a different modality often elicits dramatically different behavior from the same model. A model that easily solves a geometry problem from a text description may completely fail on the same problem presented as a diagram—not because it lacks geometric knowledge, but because it fails to activate that knowledge from visual input.

Why Existing Approaches Fall Short

Standard VLM post-training mixes image-text pairs and optimizes a single objective (SFT or RLHF) over all examples, treating modality as a property of the input rather than an independent factor in reasoning quality. This approach:

- **Averages over view-dependent performance:** A model that excels on text but fails on diagrams receives the same training gradient regardless of which view it was shown.
- **Does not exploit complementary signals:** The model's success on text implicitly encodes geometric reasoning that could guide improvement on diagrams—but standard training does not route this signal across modalities.
- **Ignores asymmetric capability:** Per-view performance profiles differ across problems; a fixed training mix cannot leverage which view each problem is easiest from.

Core Mechanism

MIRROR introduces a per-problem view evaluation step before each RL update. For each training problem p:

1. Evaluate the current policy on all views $\mathcal{V} = \{T, I, T+I\}$ to obtain per-view rewards $r(p, v)$.
2. Identify the best view: $v^*(p) = \arg\max_v r(p, v)$.
3. Compute augmented rewards for weaker views that include a cross-modal alignment term encouraging the weaker-view distribution to match the best-view distribution.

The policy update uses GRPO with the augmented rewards, simultaneously training on all views. The reciprocal structure ensures that improvement on the text view can feed into the image view and vice versa.

Technical Formulation

Let π_θ be the policy and a the correct answer. The base reward is $r(p, v) = \mathbf{1}[\pi_\theta(p_v) = a]$. The MIRROR-augmented reward for view $v \neq v^*(p)$ is:

$$\tilde{r}(p, v) = r(p, v) + \lambda \cdot \mathrm{KL}\Big(\pi_\theta(\cdot \mid p_{v^*}) \Big\| \pi_\theta(\cdot \mid p_v)\Big)^{-1}$$

The KL term penalizes divergence from the best-view response distribution, rewarding weaker views for matching the best-view reasoning trace. The inverse ensures the bonus is higher when the weaker view is already close to the best view (facilitating transfer) and smaller when the gap is large (avoiding overfit to a single view).

The full GRPO objective with MIRROR shaping:

$$\mathcal{L}_{\text{MIRROR}} = -\mathbb{E}_{p,v\sim\mathcal{D}}\left[\tilde{r}(p, v)\,\frac{\pi_\theta(a_v \mid p_v)}{\pi_{\theta_{\text{old}}}(a_v \mid p_v)} - \beta\,\mathrm{KL}(\pi_\theta \| \pi_{\text{ref}})\right]$$

Learning or Inference Procedure

Training uses ODA-Data for all views simultaneously. Each mini-batch contains all three views of the same problem, enabling within-batch comparison of view-dependent rewards. The λ coefficient governing cross-modal shaping strength is annealed from 0.5 to 0.1 over training to prevent the alignment term from dominating late-stage learning.

At inference, no change is needed: the model is queried with whichever view is available. The improvement is baked into the model weights.

What the Guarantee Says

MIRROR does not guarantee cross-view performance parity—some problems may remain fundamentally harder in one view than another. What it provides is a systematic mechanism for reducing the gap, ensuring that the model's best-view capabilities are not siloed and can transfer across the view boundary.

Experimental Findings

ODA-Data test set:

Method	Text	Image	Combined
LLaVA-1.6 (base)	61.3	44.2	58.7
+ GRPO only	68.1	49.8	64.3
+ MIRROR	**74.5**	**61.2**	**70.8**

External geometry benchmarks (average over GeoQA, UniGeo, Geometry3K): MIRROR +5.8pp over GRPO-only, +11.4pp over base.

The image-dominant view shows the largest absolute gain (+17pp over base), consistent with the hypothesis that image reasoning benefits most from cross-modal transfer from the textually stronger reasoning path.

Ablations and Interpretation

Remove cross-modal shaping ($\lambda = 0$): Image accuracy drops 9.4pp relative to full MIRROR. The model still improves over base from outcome RL, but the cross-view transfer is lost.

Best view only (train only on v^*, ignore weaker views): Image accuracy drops 14.1pp relative to MIRROR. Focusing only on the easiest view degrades weaker-view performance—the model over-specializes.

Scrambled skills (random KL target per problem): Image accuracy falls to 47.1%, below the GRPO-only baseline. The content of the cross-modal alignment target—not just the fine-tuning signal—drives the improvement.

λ annealing: Fixed $\lambda = 0.5$ degrades image accuracy by 3.2pp in late training, confirming that annealing is important for preventing the alignment term from dominating the RL signal.

Reference

Wen Ye, Yuxiao Qu, Aviral Kumar, Xuezhe Ma. **MIRROR: Learning from the Other View for Multi-Modal Reasoning**. arXiv:2607.21552, July 23, 2026. https://arxiv.org/abs/2607.21552

Keywords: diffusion models • inference-time optimization • Kalman filtering • ICML 2026

DiFA: Inference-Time Forward-Process Alignment for Diffusion Models

Treating Each Denoising Step as a Kalman Observation Yields a Training-Free Temporal Consensus Wrapper with Significant FID Gains on CIFAR-10 and ImageNet

By Shigui Li, Delu Zeng (South China University of Technology)

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Standard diffusion model inference treats each denoising step as an independent numerical integration, discarding the statistical correlation structure of predictions along the reverse trajectory. DiFA (**Di**ffusion **F**orward-process **A**lignment), accepted at ICML 2026, reframes inference as sequential state estimation inspired by Kalman filtering: each prediction is an observation of the true sample, fused with prior predictions according to noise-level compatibility to build a forward-aligned temporal consensus. The result is a training-free wrapper that wraps any existing sampler and yields consistent FID improvements.

AT A GLANCE

Problem: Inference treats denoising steps as independent estimators of x_0, ignoring forward-process statistical correlations across the trajectory.

Why it matters: Exploiting the statistical structure of the forward process during inference is free—it requires no training and can improve any existing sampler.

Method: Noise-level-weighted temporal consensus aggregates historical predictions; deviation guidance preserves fine-grained detail that averaging would smooth out.

Result: FID 4.67→3.21 on CIFAR-10 with DDIM (20 steps); FID 12.4→8.9 on ImageNet 256×256 with DDIM (50 steps).

Evidence: Ablations confirm both the consensus and deviation components are necessary; uniform averaging instead of noise-weighted degrades FID by 0.4–1.2.

The Big Picture

Diffusion models generate images by reversing a noise-injection process: starting from Gaussian noise x_T, a learned denoiser iteratively predicts the clean sample x_0 and steps toward it. The standard inference framework—DDIM, DPM-Solver, EDM—casts each step as deterministic numerical integration of the score function, treating the model's prediction $\hat{x}_0^{(t)}$ at each step as an independent, exact estimate.

DiFA identifies a missed opportunity: the sequence of predictions $\{\hat{x}_0^{(T)}, \hat{x}_0^{(T-1)}, \ldots, \hat{x}_0^{(t)}\}$ is not independent—it represents a series of increasingly refined observations of the same unknown x_0, each corrupted by noise at level $\bar{\alpha}_t$. Kalman filtering is the optimal framework for fusing such a sequence of noisy observations. DiFA imports this framework into diffusion inference.

Why Existing Approaches Fall Short

The numerical integration perspective assumes each $\hat{x}_0^{(t)}$ is an unbiased, independent estimator of x_0. In practice:

Early predictions are structurally aligned but coarse. At high noise levels t, the model has little to work with—predictions are diffuse and uncertain. Yet they are statistically consistent with the forward process, making them useful priors for later steps.

Late predictions are fine but may drift. At low t, the model makes confident, detailed predictions—but these can deviate from the statistical manifold of the forward process, leading to artifacts or mode drops. Independent step treatment cannot detect or correct this drift.

No cross-step information reuse. Current samplers discard $\hat{x}_0^{(\tau)}$ for $\tau > t$ once step t is reached, wasting information that has already been computed.

Core Mechanism

DiFA introduces **Forward-Process Alignment (FPA)**: at step t, instead of using $\hat{x}_0^{(t)}$ directly, it computes a weighted consensus $\tilde{x}_0^{(t)}$ across all predictions to date:

$$\tilde{x}_0^{(t)} = \sum_{\tau \geq t} w(\tau, t)\, \hat{x}_0^{(\tau)}, \qquad \sum_{\tau} w(\tau, t) = 1$$

Weights $w(\tau, t)$ are larger for predictions at noise levels close to t (structurally compatible) and smaller for predictions made at very different noise levels (less compatible):

$$w(\tau, t) \propto \exp\left(-\frac{\|\hat{x}_0^{(\tau)} - \hat{x}_0^{(t)}\|^2}{2\sigma^2}\right) \cdot \frac{1 - \bar{\alpha}_\tau}{1 - \bar{\alpha}_t}$$

To prevent over-smoothing, DiFA adds **Deviation Guidance**:

$$\hat{x}_0^{\mathrm{DiFA}}(t) = \tilde{x}_0^{(t)} + \gamma\left(\hat{x}_0^{(t)} - \tilde{x}_0^{(t)}\right)$$

where $\gamma \in (0, 1)$ controls the trade-off between consensus smoothness and single-step detail.

Technical Formulation

The forward process is:

$$q(x_t \mid x_0) = \mathcal{N}\big(x_t;\, \sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t)I\big)$$

Treating each $\hat{x}_0^{(\tau)}$ as an observation of x_0 with additive Gaussian noise of variance proportional to $(1-\bar{\alpha}_\tau)$, the minimum-variance linear estimator is:

$$\tilde{x}_0^{(t)} = \left(\sum_{\tau \geq t} \frac{1}{1-\bar{\alpha}_\tau}\right)^{-1} \sum_{\tau \geq t} \frac{\hat{x}_0^{(\tau)}}{1-\bar{\alpha}_\tau}$$

DiFA approximates this by replacing the Gaussian noise assumption with the empirical prediction discrepancy, giving the noise-and-consistency weighted formula above. The two formulations agree when predictions are unbiased; DiFA's version is robust to the bias introduced by imperfect score estimation.

Learning or Inference Procedure

DiFA requires no additional training or fine-tuning. The implementation wraps the inner loop of any existing sampler:

1. Run the sampler's score evaluation to obtain $\hat{x}_0^{(t)}$.
2. Append $\hat{x}_0^{(t)}$ to the history buffer.
3. Compute $\tilde{x}_0^{(t)}$ via noise-level-weighted averaging over the buffer.
4. Apply deviation guidance to obtain $\hat{x}_0^{\mathrm{DiFA}}(t)$.
5. Pass $\hat{x}_0^{\mathrm{DiFA}}(t)$ to the sampler's step function in place of $\hat{x}_0^{(t)}$.

Memory overhead scales as $O(T \cdot H \cdot W \cdot C)$ for the history buffer; at 50 steps and 256×256 resolution, this is approximately 800 MB, which fits on a standard GPU.

What the Guarantee Says

Under Gaussian noise assumptions, DiFA's temporal consensus is the minimum-variance linear estimator of x_0 given all observations. This provides an optimality guarantee for the consensus itself.

For the full DiFA procedure (consensus + deviation guidance), the optimality guarantee applies in expectation to the forward-aligned component; the deviation term is a practical correction for the non-Gaussian case, empirically validated but without closed-form optimality.

Experimental Findings

CIFAR-10 unconditional (FID ↓):

Sampler	Steps	Baseline	+DiFA
DDIM	20	4.67	**3.21**
DPM-Solver++	20	3.44	**2.87**
EDM	35	2.04	**1.78**

ImageNet 256×256 class-conditional (FID ↓):

Sampler	Steps	Baseline	+DiFA
DDIM	50	12.4	**8.9**
DPM-Solver++	50	7.8	**5.6**

FD-DINOv2 (perceptual quality metric) and Inception Score improve consistently across all configurations. The largest absolute FID improvement occurs with DDIM, which makes independent step predictions with no built-in cross-step information reuse—confirming that DiFA fills a gap most pronounced in first-order samplers.

Ablations and Interpretation

Pure consensus ($\gamma = 0$): FID improves but visual sharpness drops; generated samples appear over-smoothed compared to reference images.

No consensus ($\gamma = 1$): Equivalent to baseline. The consensus is doing the work.

Full history vs. last 5 steps: Using the full prediction history improves FID by an additional 0.3–0.8 points, confirming that early-trajectory predictions carry useful structural information even when fused with weights that naturally down-weight them.

Uniform averaging vs. noise-level weighting: Replacing the noise-level compatibility term with uniform averaging degrades FID by 0.4–1.2 points. Noise-level weighting is critical: predictions made at very different noise levels have structurally incompatible biases that uniform averaging cannot account for.

Computational overhead: Adding DiFA to DDIM (20 steps) on a single A100 increases wall-clock inference time by 6.4%—the dominant cost is the buffer read and weighted average, not additional neural function evaluations.

Reference

Shigui Li, Delu Zeng. **DiFA: Inference-Time Forward-Process Alignment for Diffusion Models**. arXiv:2607.17972. ICML 2026. July 19, 2026. https://arxiv.org/abs/2607.17972

Keywords: agentic reinforcement learning • hindsight distillation • LLM agents • on-policy learning

SEED: Self-Evolving On-Policy Distillation for Agentic Reinforcement Learning

Converting On-Policy Trajectories into Hindsight Skills and Distilling Them Back into the Policy Closes the Credit Assignment Gap in Long-Horizon Agentic Tasks

By Jinyang Wu, Shuo Yang, Zhengxi Lu, Fan Zhang, Yuhao Shen, Lang Feng, Haoran Luo, Zheng Lian, Shuai Zhang, Zhengqi Wen, Jianhua Tao (University of Science and Technology of China)

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Outcome-based reinforcement learning for long-horizon agentic tasks provides only a single reward signal at the end of trajectories spanning 20–50 decision steps, leaving a large credit assignment gap at the token level. SEED fills this gap by converting completed on-policy trajectories into natural-language hindsight skills—reusable workflows, decisive observations, failure-avoidance rules—and distilling them back into the same policy that generated the trajectories. The policy and its hindsight supervisor co-evolve across rounds, outperforming both pure RL and static distillation baselines on AlfWorld and WebArena.

AT A GLANCE

Problem: Sparse outcome rewards in long-horizon agentic RL provide insufficient token-level supervision, leaving the policy to learn intermediate decision quality from a single scalar.

Why it matters: Better intermediate supervision enables smaller models to learn complex agentic tasks—SEED yields the largest gains at 3B parameters.

Method: The policy analyzes its own completed trajectories into hindsight skills, then is distilled on (x, skill, a) triplets; the loop repeats as the policy evolves.

Result: AlfWorld success rate 74.2% (7B) vs. 72.8% GRPO-only; WebArena 26.3% vs. 18.4% GRPO-only.

Evidence: Fixed analyzer loses 5.1pp; scrambled skills fall below GRPO-only, confirming hindsight content drives gains.

The Big Picture

Training LLMs to act as autonomous agents—navigating web UIs, executing shell commands, managing files across multi-step workflows— requires learning from environmental feedback over trajectories that may span dozens of decisions. GRPO and similar outcome-based RL methods assign the same reward to every token in the trajectory based on the final outcome, providing no signal about which intermediate actions were decisive.

SEED treats the policy as its own supervisor: after completing a trajectory, the same model reads it and generates a structured retrospective capturing what worked, what failed, and what it should do differently—then trains on this signal. When the policy improves, its next-round retrospectives become more accurate, creating a self-evolving supervision signal that stays aligned with the model throughout training.

Why Existing Approaches Fall Short

Pure outcome RL (GRPO): Gradient propagates from the final reward backward through all steps. Effective action-level attribution requires the model to implicitly learn which actions contributed to the outcome—a difficult credit assignment problem over long trajectories.

Static distillation: A stronger model (teacher) generates demonstrations or skills, and the student is trained on them. The teacher is fixed: once the student surpasses the teacher on some tasks, the distilled supervision becomes stale and may even be harmful.

Off-policy hindsight: Learning from past trajectories (collected by earlier policy versions) introduces distribution shift between the generating policy and the current policy. Skills derived from an earlier policy's behavior may not apply to the current policy's failure modes.

Core Mechanism

SEED runs three operations using the same model checkpoint, synchronized in a single training round:

1. **Act:** Policy π_k rolls out trajectories $\{\tau_i\}$ in the environment.
2. **Analyze:** Policy π_k reads each completed τ_i and generates a natural-language hindsight skill s_i: a 2–4 sentence description of the key decision pattern, crucial observation, or failure cause in that trajectory.
3. **Distill:** Policy π_{k+1} is trained on (x_i, s_i, a_i) triplets where x_i is the task context, s_i is the hindsight skill, and a_i is the correct action.

After round k, π_{k+1} becomes the actor and analyzer for round $k + 1$. Skills generated by π_{k+1} reflect its own current capabilities and remaining failure modes.

Technical Formulation

Let $\tau = (o_1, a_1, \dots, o_T, a_T, R)$ be a trajectory. The hindsight skill generation prompt $\mathcal{P}_{\text{analyze}}(\tau)$ instructs the model to identify: (a) which intermediate action most determined the outcome, (b) which environmental observation was the pivotal evidence, and (c) which alternative action would have changed the result.

The skill generation objective:

$$s_i = \pi_k(\cdot \mid \mathcal{P}_{\text{analyze}}(\tau_i))$$

The distillation objective (behavior cloning with skill context):

$$\mathcal{L}_{\text{SEED}}(\theta) = -\mathbb{E}_{(x,s,a)\sim\mathcal{D}_k}[\log \pi_\theta(a \mid x, s)]$$

The full training objective mixes outcome RL and skill distillation:

$$\mathcal{L}_{\text{total}} = (1-\alpha)\,\mathcal{L}_{\text{GRPO}} + \alpha\,\mathcal{L}_{\text{SEED}}$$

with $\alpha = 0.3$ in all experiments. GRPO provides the global policy gradient; SEED provides the local, token-level supervision.

Learning or Inference Procedure

Each SEED round proceeds as:

1. Sample B trajectories from π_k in parallel.
2. For each trajectory, run the analysis prompt to obtain s_i.
3. Construct the distillation dataset $\mathcal{D}_k$.
4. Train π_{k+1} on $\mathcal{L}_{\text{total}}$ for one epoch.

At inference, the model is queried without any hindsight skill context; skills are a training-time scaffold only, not an inference-time input.

What the Guarantee Says

SEED does not guarantee that hindsight skills are accurate—the model may generate incorrect retrospectives about its own behavior. What it does guarantee is *alignment*: since the analyzer and actor share weights, the analysis is always conducted from the perspective of the current policy's capabilities. As the policy improves, its analyses improve with it. This is a softer guarantee than formal credit assignment—but it holds without any external supervision.

Experimental Findings

AlfWorld (household task success ↑):

Method	3B	7B
GRPO only	61.3	72.8
Static distillation	67.4	78.1
SEED (ours)	**74.2**	**83.5**

WebArena (task success ↑):

Method	7B
GRPO only	18.4
Static distillation	21.7
SEED (ours)	**26.3**

The 3B model shows the largest proportional gain (+13pp on AlfWorld vs. GRPO-only), confirming that explicit hindsight supervision is most valuable when model capacity is limited.

Ablations and Interpretation

Fixed analyzer (π_0 as analyzer throughout): AlfWorld accuracy drops 5.1pp relative to full SEED. Stale analyzers produce skills that reflect initial failure modes, not current ones.

Scrambled skills (random pairing of skill s_j with context x_i): Accuracy falls to 59.8% on AlfWorld—below GRPO-only at 61.3%. Incorrect skills actively hurt performance, confirming that content quality, not training format, drives the improvement.

Analysis without distillation (GRPO + analysis as internal chain-of-thought): Improves AlfWorld by 1.4pp over GRPO-only. The self-analysis provides modest benefit as a form of implicit chain-of-thought, but the formal distillation step is responsible for the majority of gains.

α **sensitivity:** $\alpha \in [0.2, 0.4]$ gives similar results. $\alpha > 0.6$ degrades GRPO performance as distillation dominates.

Reference

Jinyang Wu, Shuo Yang, Zhengxi Lu, Fan Zhang, Yuhao Shen, Lang Feng, Haoran Luo, Zheng Lian, Shuai Zhang, Zhengqi Wen, Jianhua Tao. **SEED: Self-Evolving On-Policy Distillation for Agentic Reinforcement Learning**. arXiv:2607.14777, July 16, 2026. https://arxiv.org/abs/2607.14777

Keywords: vision-language-action models • robot learning • 3D pointmaps • viewpoint generalization

See like a Robot: Robot-Centric Pointmaps for Vision-Language-Action Models

Replacing Camera-Frame RGB Images with Robot-Frame 3D Pointmaps Closes the Viewpoint Generalization Gap in Large-Scale Robot Learning Datasets

By Byungkun Lee, Dongyoon Hwang, Dongjin Kim, Hojoon Lee, Minho Park, Jaegul Choo (KAIST AI; Holiday Robotics)

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Vision-language-action (VLA) models predict robot actions defined in the robot's own 3D body frame but observe the world through camera images defined in the camera's frame—a mismatch that is benign under fixed cameras but harmful when training datasets aggregate demonstrations from diverse camera setups. Researchers at KAIST AI and Holiday Robotics propose **robot-centric pointmaps**: images whose pixels store the 3D position of scene points in the robot frame, dropping straight into existing VLA architectures with a single channel-wise concatenation. Cross-camera evaluation improves by 20 percentage points over RGB-only baselines.

AT A GLANCE

Problem: VLA models observe scenes in the camera frame but act in the robot frame; diverse camera setups in large datasets make this frame mismatch a primary generalization barrier.

Why it matters: Large-scale robot learning requires diverse data from many camera rigs; solving viewpoint generalization unlocks the full benefit of aggregated datasets.

Method: Pointmaps—H×W images with pixel values equal to the 3D robot-frame position—are concatenated with RGB and passed to an existing VLA with one adapted input layer.

Result: Cross-camera success 61.4% vs. 41.2% for RGB only on Open X-Embodiment; +20–30pp on real-robot tasks.

Evidence: Camera-frame 3D gives only 51.3% cross-camera success, confirming the coordinate system (not just 3D info) is the key factor.

The Big Picture

Scaling robot learning with broad, diverse datasets—analogous to what large language models did with internet text—is a central goal in robotics. Datasets like Open X-Embodiment aggregate millions of demonstrations from dozens of robot platforms and camera setups. But unlike text, where position in the document is largely irrelevant to the downstream task, in robot learning the camera viewpoint fundamentally changes the appearance of the same scene.

A VLA trained on overhead camera demonstrations cannot directly generalize to wrist-camera deployment because the pixel-to-action mapping changes completely. If the policy could observe the scene in the robot's own coordinate frame, viewpoint would become irrelevant: the spatial relationship between the robot and the object is the same regardless of which camera captured it.

Why Existing Approaches Fall Short

RGB-only VLAs: Learn to infer spatial relationships from monocular cues—perspective foreshortening, shadows, texture gradients. These cues are camera-dependent and do not generalize across camera setups without explicit camera-pose conditioning.

Camera-frame depth augmentation: Providing depth in the camera frame adds metric distance information but not robot-frame position. The mapping from camera-frame depth to robot-frame coordinates still requires extrinsic calibration knowledge—which may differ across demonstrations in a diverse dataset.

Camera-pose conditioning: Explicitly providing camera extrinsics as input to the policy has been explored, but requires accurate calibration at deployment and adds architectural complexity. Policies that memorize pose→action mappings may overfit to training camera configurations.

Core Mechanism

A **robot-centric pointmap** $P \in \mathbb{R}^{H\times W\times 3}$ is an image where each pixel (u, v) stores the 3D coordinate (X, Y, Z) of the corresponding scene point in the robot's body frame. The pointmap is constructed at dataset-collection time:

1. Backproject each pixel to 3D in the camera frame using depth $d_{u,v}$ and intrinsic matrix K.
2. Transform to the robot frame using the known extrinsic $T_{\text{cam}} \to T_{\text{robot}}$.
3. Store the result as the pixel value.

The policy receives $[I_{\text{RGB}}, P] \in \mathbb{R}^{H\times W\times 6}$ as input. The visual encoder's first layer is adapted to accept 6 channels (the 3 new channels initialized to zero weights); all other weights are kept from pretrained initialization.

Technical Formulation

For pixel (u, v) with depth $d_{u,v}$, the backprojection is:

$$\mathbf{p}_{\text{cam}} = d_{u,v}\, K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

Transformation to the robot frame via rigid-body transform $T = (R, \mathbf{t})$:

$$\mathbf{p}_{\text{robot}} = R\, \mathbf{p}_{\text{cam}} + \mathbf{t}$$

The pointmap entry is $P_{u,v} = \mathbf{p}_{\text{robot}}$. This computation runs at over 60 Hz on standard GPU hardware.

For datasets where depth sensors are unavailable, the authors use Depth Anything V2 (monocular estimator) followed by scale-and-shift alignment using known anchor points (e.g., robot end-effector position). Monocular pointmaps lag behind sensor-depth pointmaps by only 4.2pp on cross-camera evaluation, showing practical viability without depth hardware.

Learning or Inference Procedure

Training follows the standard VLA pipeline—cross-entropy on action tokens given visual and language input—with the single change of replacing RGB-only input with 6-channel RGB+pointmap input. No changes to the loss, optimizer, tokenizer, or action discretization are required.

At deployment, the robot's camera calibration (intrinsics and extrinsics relative to its body) is used to compute pointmaps in real time. The model is never given raw camera-frame depth or explicit extrinsics as an input token; the coordinate transformation is baked into the pointmap pixels.

What the Guarantee Says

If the camera-to-robot transform T is known exactly and the depth estimate $d_{u,v}$ is accurate, the pointmap provides exact robot-frame coordinates. Under this assumption, two demonstrations of the same robot-object configuration from different cameras produce identical pointmaps—guaranteeing that the policy input is camera-invariant.

In practice, depth estimation noise and calibration error introduce per-pixel errors in the pointmap. The empirical results show that the policy is robust to these errors up to sensor-depth accuracy levels.

Experimental Findings

Open X-Embodiment cross-camera (task success $\uparrow$):

Model	Same-cam.	Cross-cam.
π_0 (RGB only)	74.3	41.2
π_0 + depth (cam. frame)	77.1	46.8
π_0 + pointmap (robot frame)	**80.6**	**61.4**

Real-robot manipulation (50 trials each):

Task	RGB only	+ Pointmap
Pick (varied height)	42%	72%
Place (varied surface)	58%	84%
Reach past obstacle	34%	64%

The largest absolute gains appear in tasks requiring precise distance judgment—picking from varying heights and navigating reach obstacles—consistent with the hypothesis that robot-frame metric information is the primary missing signal.

Ablations and Interpretation

Camera-frame vs. robot-frame 3D: Using depth in the camera frame yields cross-camera success of 46.8% vs. 61.4% for robot-frame pointmaps. 3D information helps only when it is in the robot's native coordinate system.

Monocular vs. sensor depth: Sensor depth yields 61.4% vs. 57.2% for monocular depth, a 4.2pp gap. Monocular estimation is a viable substitute in deployment environments where depth sensors are unavailable.

Channel fusion vs. early fusion: Channel concatenation (6-channel input) outperforms early-layer feature fusion (separate RGB and 3D branches merged at layer 3) by 2.8pp. Direct concatenation preserves the ability to use pretrained visual features on RGB while accessing 3D separately.

Effect of dataset diversity: Training with 10 camera configurations (vs. 2) improves cross-camera accuracy by 8.6pp for RGB-only but only 2.1pp for pointmap-based models. Pointmaps substantially reduce the benefit of camera-diverse training data, confirming that robot-frame grounding eliminates most of the viewpoint variance.

Reference

Byungkun Lee, Dongyoon Hwang, Dongjin Kim, Hojoon Lee, Minho Park, Jaegul Choo. **See like a Robot: Robot-Centric Pointmaps for Vision-Language-Action Models**. arXiv:2607.11498, July 13, 2026. https://arxiv.org/abs/2607.11498